Complementarity-Aware Local–Global Feature Fusion Network for Building Extraction in Remote Sensing Images

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***Abstract*— Building extraction is a challenging research direction in remote sensing image (RSI) interpretation. Due to the fact that a building has not only its own local structures but also similar architectural styles with other buildings located in a global area (e.g., street or community), fusing local and global features becomes a promising way to improve the performance of building extraction. Focused on this, we propose a new complementarity-aware local–global feature fusion network (CLGFF-Net) by integrating a convolutional branch and a Transformer branch. The two branches, respectively, capture local patterns and global long-range dependencies of RSIs, thereby leading to highly complementary features. To dig out the implicit complementary information for fusion, we develop a complementarity-aware fusion module (CFM) that separates shared features (SFs) and distinct features (DFs) between two branches, by building a commonalities analysis path and two difference analysis paths. Meanwhile, to make sure the similarity of SFs and dissimilarity of DFs, a triplet loss function is designed to enforce the distances between SFs to be near and DFs to be far. By this way, complementary information can be explicitly included in DFs and is adaptively exchanged between two branches for fusion. In addition, since multilayer features in each branch generally convey different-level semantic information, a multilayer fusion scheme (MLFS) is designed to fuse them by introducing cross-layer connections and gate mechanism. By coupling CFMs with MLFS, the abilities in characterizing local and global context information, as well as different-level semantic information, can be fully exploited for better mapping of complicated building objects. The experimental results demonstrate the effectiveness of our proposed method.**

***Index Terms*— Building extraction, complementary information, convolutional branch, feature fusion, Transformer branch (TFB).**

I. Introduction

BUILDING extraction in remote sensing images (RSIs) aims at assigning a class label ("building" or "nonbuilding") to each pixel in the image and has many practical applications, such as urban planning, economic assessment, and land-use mapping [1], [2], [3]. As the resolution of RSI increases, building extraction faces the challenges of complex backgrounds and diverse building structures. Early related works attempted to extract handcrafted building features (e.g., shape [4], edge [5], and shadow [6]) for binary classification [7], [8]. These features are generally recognized as low-level features and seriously rely on prior experience and expert knowledge. To adaptively learn high-level features from numerous data, the deep learning (DL) technique is also introduced in building extraction, which has presented promising performance and becomes a hot topic at present.

Among the DL methods, convolutional neural network (CNN) is one powerful feature learning model in different computer vision tasks. For the past years, various convolutional networks for semantic segmentation have been proposed and good performance is achieved, e.g., PSPNet [11], DeepLabv3+ [12], HRNet [13], and ConvNeXt [14]. Due to the task similarity between semantic segmentation and building extraction, one natural thinking to achieve building extraction task is the utilization of semantic segmentation networks. In addition, regarding the specific characters of building extraction, another branch of methods relies on applying and modifying the fully convolutional network (FCN) [15]. The good property of FCN is that it can flexibly accept input images of arbitrary sizes and efficiently generate correspondingly-sized outputs. For instance, Ronneberger et al. [16] used skip connections to build the information flow between related low-level and high-level feature maps, to exploit the structural information of the shallow features. Zhu et al. [17] further proposed a multiparallel path network structure, which can achieve efficient fusion of high-/low-resolution features, exhibiting strong semantic representation of buildings and backgrounds.

Another direction to learn more discriminative local spatial feature representation is to revise the vanilla convolution, resulting in better building extraction performance. For example, Yu et al. [18] proposed a dilated convolution, which expanded the receptive field without reducing the spatial resolution of the features, allowing the network to capture a wider range of contextual information. Dai et al. [19] introduced 2-D offsets to the regular grid sampling locations in the standard convolution, which can better extract features of objects with different shapes. Moreover, some people focused on the few-shot segmentation tasks. For instance,

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Yao et al. [9] proposed a scale-aware few-shot semantic segmentation (FSS) network for aerial images, which utilizes multiple prototypes for detail matching and exhibits superior segmentation in aerial images. Li et al. [10] innovatively introduced confidence-iou collaborative proposal filtration scheme and tiny object constraint loss to address the challenges in few-shot aerial image object detection. In conclusion, these CNN-based methods tend to capture local patterns and lack the ability to model global long-range dependencies [20], making it difficult to completely and accurately extract buildings from complex backgrounds.

To alleviate this problem, some mechanisms, e.g., attention and uncertainty prediction, are, respectively, introduced in the field of building extraction. For instance, Bello et al. [21] proposed to augment convolutional operators with self-attention mechanism by concatenating convolutional feature maps with a set of feature maps produced via self-attention. Pan et al. [22] decomposed the operations of both convolution and self-attention and then designed a hybrid operator to combine them. Moreover, He et al. [23] proposed an uncertainty-aware network (UANet), which introduces the concept of uncertainty for building extraction. Compared with some existing methods, UANet makes the network more focused on uncertain pixels by modeling the background uncertainty and building uncertainty in RSIs. As a result, when dealing with challenging scenarios, especially around building edges, UANet demonstrates improved performance.

In addition to CNN models, Transformer have recently emerged as another hot-spot DL model and gained much attention [24], [25], [26], [27], [28]. These methods adopt a self-attention mechanism to generate feature maps, which do not have the same spatial constraints as convolution calculations. Thus, it can handle a larger receptive field than conventional convolution and can effectively obtain global information. Thanks to such good characteristics, some successful attempts have been done to apply Transformer to distinguish between buildings and backgrounds in complex scenes (e.g., urban areas) [29], [30], [31]. Chen et al. [32] first used Transformer to model global long-range dependencies in both spatial and channel dimensions. To further characterize building features at different scales in RSIs, Zhang et al. [33] adopted Swin-Transformer [26] as the backbone to effectively extract multiscale features. Recently, some scholars tried to introduce locality to Transformer to better focus on the changeable shapes and dense localizations of buildings in RSIs. For instance, Wu et al. [34] incorporated the convolution into the Transformer block, enabling the network to have the benefits of both the convolution and the Transformer. Wang et al. [35] constructed a network with a two-path structure that is intended to capture global context while preserving spatial-detailed features. However, in order to process 2-D image with Transformer, the input image is first converted into a sequence of embedding vectors that correspond to the patches in the image. The self-attention mechanism then attends to all the embedding vectors and computes a weighted sum. Thus, one main issue of the Transformer is that it lacks a locality mechanism for information exchange within a single image patch, as pointed out in [36] and [37].

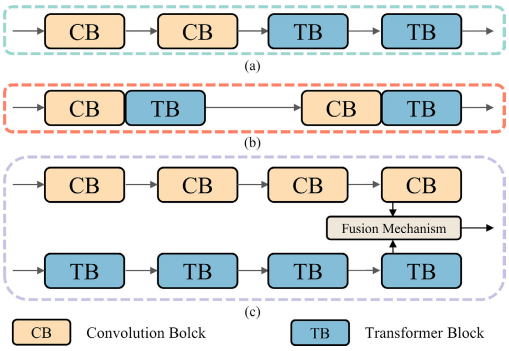


Fig. 1. Comparison of three hybrid strategies. (a) Using the convolution block in the early stages, and the Transformer block in the later stages. (b) Combining convolutional and Transformer layers to build a hybrid structure. (c) Designing a fusion mechanism to integrate the convolutional and Transformer features.

More lately, considering CNN and Transformer are obviously complementary, an effective way is to fuse both networks to improve building extraction performance. By designing effective fusion strategies, the superiority of both networks can be jointly integrated, i.e., CNN prefers to capture local patterns and the Transformer tends to model long-range global dependencies. One fusion approach [38], [39], [40] [Fig. 1(a)] adopted convolution blocks in the shallow stages, followed by the stacking of Transformer blocks in the deep stages. However, this strategy fails to capture global context information in the shallow stages, which is crucial for the performance of dense prediction tasks (e.g., semantic segmentation). Another fusion approach [27], [28] [Fig. 1(b)] aims to combine convolutional layers with multihead attention layers to form a new structure. For example, the convolutional Transformer block in convolutional vision transformer (CVT) [34] consists of a depthwise separable convolutional layer and a multihead attention layer that can capture both local and global information, resulting in powerful modeling capabilities. The third fusion strategy [29], [30] [Fig. 1(c)] attempted to use a parallel network architecture, i.e., the CNN branch (CVB) and Transformer branch (TFB), to learn local and global features separately and fuse them later. It allows to design various fusion mechanisms (e.g., feature-level fusion and decision-level fusion) based on specific tasks, to fully utilize the characteristics of each branch. Therefore, we adopt the third fusion strategy in this work, while need to deal with two inevitable problems, i.e., what to fuse (WOF) and how to fuse (HOF).

In order to solve the above issues, a complementarity-aware local–global feature fusion network (CLGFF-Net) is proposed in this article. For WOF, we attempted to deeply explore the complementary information in CNN and Transformer networks, and then use those information as well as complementary multilayer features for fusion. For HOF, we construct a two-branch structure using CNN and Transformer backbones and build a feature fusion scheme which coupling the interbranch fusion and intrabranch fusion. In specific, a novel complementarity-aware fusion module (CFM) and multilayer fusion scheme (MLFS) are designed to, respectively, fuse interbranch and intrabranch features. As a consequence, local and global context information from interbranch features as well as semantic information of different

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layers from intrabranch features are fully utilized to improve the performance of building extraction in RSIs. The main contributions of this work are listed as follows.

1) A new end-to-end building extraction network, which focuses on fusing complementary CVB and TFB, is developed. Benefiting from the well-designed fusion scheme, the network allows capturing both local patterns and global long-range dependencies in RSIs, thus enhancing the ability of characterizing local and global spatial relation cues.
2) A novel CFM is proposed to fuse interbranch features. On the one hand, a commonalities analysis path (CAP) and two difference analysis paths (DAPs) are built with the regularization of a designed triplet loss function, to separate shared features (SFs) and distinct features (DFs) between branches. On the other hand, DFs are fused in a cross manner to adaptively exchange complementary information with a developed gate unit.
3) To fuse intrabranch features, the MLFS is developed to build multiple cross-layer connections within each branch, which facilitate the integration of semantic information from different layers. By leveraging MLFS, we can obtain feature representations with rich multilevel semantic information, enabling better mapping of complex building objects.

The rest of this article is structured as follows. Some related works are briefly introduced in Section II. Section III introduces the proposed network in detail. In Section IV, we report the experimental results and provide corresponding analysis. Finally, this work is summarized in Section V.

II. Related Work

Feature extraction is critical for the building extraction task. Two well-known deep feature learning models, i.e., the CNN and Transformer, are simultaneously used as backbones in the proposed method. In this section, we give a brief overview of the very deep convolutional network [Visual Geometry Group (VGG)] [41] and the vision transformer (ViT) [25] models that are, respectively, the classical CNN and Transformer networks.

A. CNN-Based Feature Extraction

The VGG network consists of multiple blocks. As shown in Fig. 2(a), each VGG block integrates multiple convolution layers and a max pooling layer. In addition, to accelerate the training of the network, we introduce the batch normalization operation for each convolution layer. The lth convolutional layer within the ith block can be represented by

$$\mathbf{Z}_i^l = \mathrm{ReLU}\left(\mathrm{BN}\left(W_l\left(\mathbf{Z}_i^{l-1}\right)\right)\right) \tag{1}$$

where $\mathbf{Z}_i^l$ and $\mathbf{Z}_i^{l-1}$ are, respectively, the output and input feature maps of ith VGG block, W_l denotes the convolution operation, BN is the batch normalization operation, and rectified linear unit (ReLU) is the ReLU activation function. In general, the number of convolution layers varies for different VGG blocks. That is, the first two VGG blocks contain two convolution layers, while other blocks have three convolution layers. By cascading VGG blocks in sequence, the VGG network can better learn features, especially for local features. These learned features help to the prediction of dense building/nonbuilding labels.

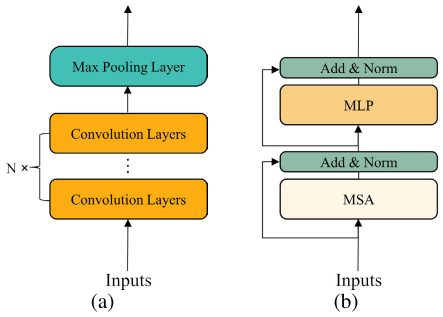


Fig. 2. Schematics of a VGG block and a transformer block. (a) VGG block. (b) ViT block.

B. Transformer-Based Feature Extraction

The ViT network is mainly composed by Transformer blocks. As shown in Fig. 2(b), each block contains a multihead self-attention (MSA) layer, a multilayer perceptron (MLP) layer, the skip connection architectures, and the layer normalization operations. Let $\mathbf{E}_{l-1} \in \mathbb{R}^{N\times d}$ and $\mathbf{E}_l \in \mathbb{R}^{N\times d}$, respectively, represent the input and output features (both of them are sequence data with length N and dimension d) of lth Transformer block. The features $\mathbf{E}_l$ are learned by

$$\begin{aligned}\mathbf{A}_{l-1} &= \mathrm{MSA}(\mathrm{LN}(\mathbf{E}_{l-1})) + \mathbf{E}_{l-1}\\ \mathbf{E}_l &= \mathrm{MLP}(\mathrm{LN}(\mathbf{A}_{l-1})) + \mathbf{A}_{l-1}\end{aligned} \tag{2}$$

where LN means the layer normalization operation, and $\mathbf{A}_{l-1}$ is the intermediate variable. The MSA layer linearly projects the normalized $\mathbf{E}_{l-1}$ to $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ of h heads and uses the self-attention mechanism to capture long-range dependencies. The calculation of MSA layer is achieved by

$$\begin{aligned}&\mathrm{MSA}(\mathrm{LN}(\mathbf{E}_{l-1})) = \mathrm{Concat}(\mathbf{S}_1\mathbf{V}_1, \mathbf{S}_2\mathbf{V}_2, \ldots, \mathbf{S}_h\mathbf{V}_h)W^O\\ &\text{s.t.}\quad \mathbf{S}_h = \mathrm{Softmax}\left(\frac{\mathbf{Q}_h\cdot\mathbf{K}_h^{\mathrm{T}}}{\sqrt{\varepsilon}}\right)\end{aligned} \tag{3}$$

where $\mathbf{Q}_h$, $\mathbf{K}_h$, and $\mathbf{V}_h$ denote the projections of normalized $\mathbf{E}_{l-1}$ in the hth head, $\mathbf{S}_h$ is the corresponding attention score, ε is the scale parameter, and W^O is the feed-forward layer that maps the concatenated features to the output space. By stacking multiple Transformer blocks, the ViT owns a powerful ability for global feature learning.

III. Proposed Method

The proposed CLGFF-Net first adopts the two-branch structure to extract multilevel local and global features. Then, these features are effectively fused by a developed CFM and multilayer feature fusion scheme. Finally, the optimization of the CLGFF-Net is achieved by designing a joint loss function. Fig. 3 demonstrates the framework of the CLGFF-Net. The detailed descriptions of the CLGFF-Net are given as follows.

A. Two-Branch Feature Extraction

In general, buildings have unique characteristics that convey discriminative local information (e.g., boundary and texture) in RSIs. In addition, buildings located in a global range (street,

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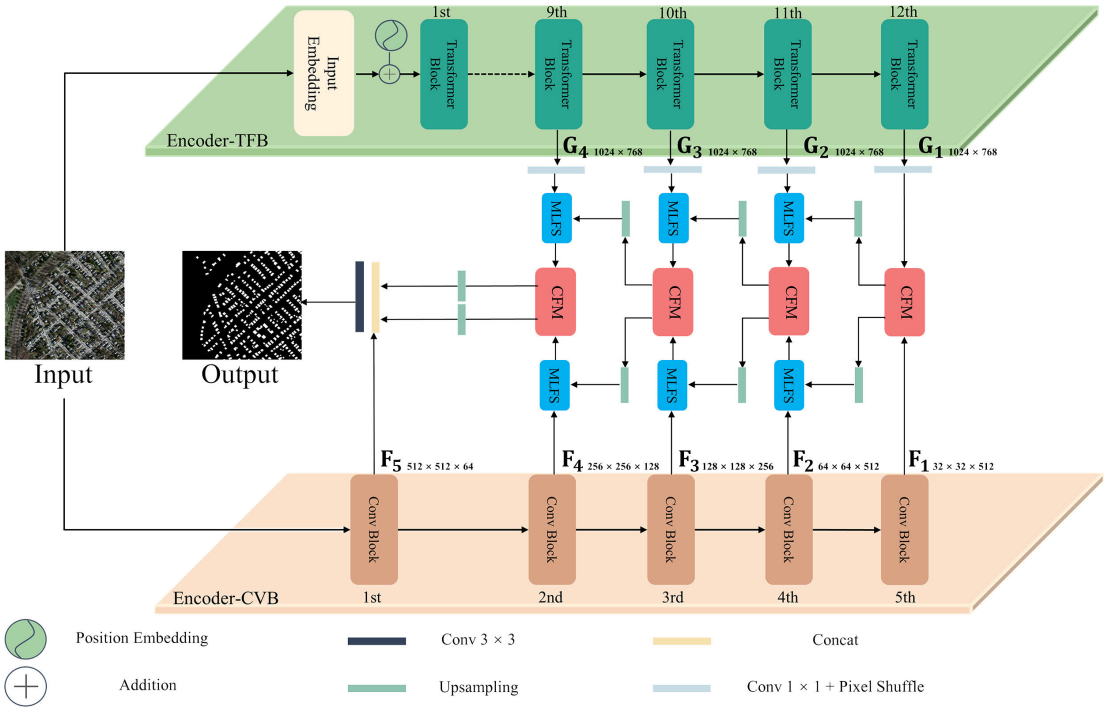


Fig. 3. Overall diagram of CLGFF-Net.

community, or city) tend to share some common characteristics (e.g., shape and color). These characteristics are distinguished from nonbuilding targets and convey discriminative global information. Therefore, it is necessary for building extraction to simultaneously leveraging the local and global context information.

To this end, we use the VGG-16 and ViT-Base to build a two-branch structure for, respectively, capturing local patterns and global long-range dependencies. Specifically, the VGG-16 network consists of five convolutional blocks by discarding all the fully connected layers, while the ViT-Base network contains 12 Transformer blocks. The former and the latter are, respectively, called the CVB and the TFB. For the CVB, we input the original image $\mathbf{X}$ and obtain five features $\{\mathbf{Z}_i\}_{i=1,2,\ldots,5}$, respectively, derived from five VGG blocks. In the following description, we use $\mathbf{F}_i = \mathbf{Z}_{6-i}$ for better representation of CNN features. For the TFB, the sequence data $\mathbf{E}_0 \in \mathbb{R}^{N\times d}$ of embedded image patches is used as the primary input signals and is calculated by

$$\mathbf{E}_0 = T(\mathbf{X}) + \mathbf{P} \tag{4}$$

where $T(\mathbf{X})$ means the operation of dividing image $\mathbf{X}$ into nonoverlapping patches and constructing the embedding vector, and $\mathbf{P}$ represents the position embedding vector. After the input of $\mathbf{E}_0$, 12 Transformer blocks output a set of features $\{\mathbf{E}_l\}_{l=1,2,\ldots,12}$ according to (2). In the following description, we use $\mathbf{G}_j = \mathbf{E}_{13-j}$ to represent TFB features. To promote the subsequent fusion of VGG and Transformer features, for $\forall i, j$, we make $\mathbf{G}_j \in \mathbb{R}^{N\times d}$ align to $\mathbf{F}_i \in \mathbb{R}^{h\times w\times c}$ by

$$\mathbf{G}_j = \text{upsampling}\big(W_{1\times 1}\big(\text{reshape}\big(\mathbf{G}_j\big)\big)\big) \tag{5}$$

where $\mathbf{G}_j$ is reshaped from 2-D to 3-D and relies on 1×1 convolution $W_{1\times 1}$ and upsampling operation to adjust feature size. As a result, the alignment of CNN features and Transformer features can be achieved.

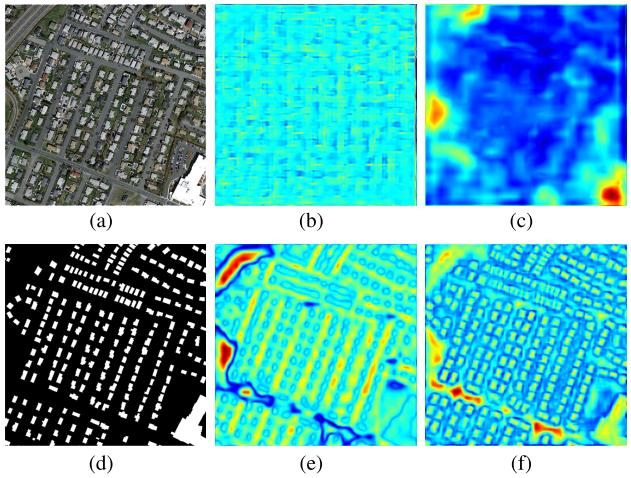


Fig. 4. Visualization of the output features from different transformer blocks. (a) Original image. (b) Output of the first transformer block. (c) Output of the second transformer block. (d) Ground truth. (e) Output of the 11th transformer block. (f) Output of the 12th transformer block.

B. Local–Global Feature Fusion

To make full use of local and global features, we develop a well-designed feature-level fusion scheme that settles two important issues, i.e., WOF and HOF. We consider the WOF in two aspects. First, it is known that CVB and TFB are strongly complementary for the extraction of local and global features. Second, many works [13], leveraging multilayer features for better performance, confirm that features of different layers are partly complementary. Therefore, interbranch and intrabranch complementarities are used for fusion.

For the HOF, a well-designed feature-level fusion scheme is developed by coupling the brand-new CFM and the MLFS.

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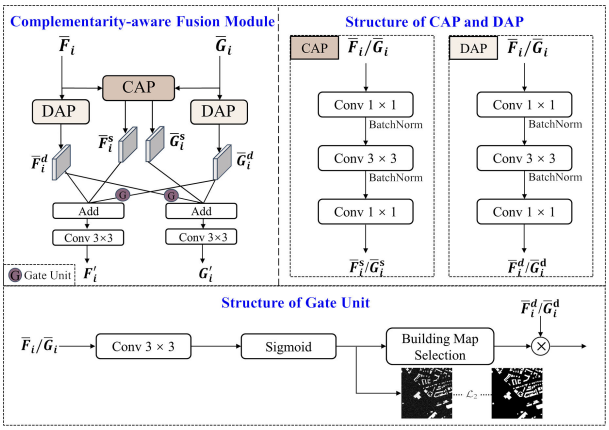


Fig. 5. Structure of a CFM, including the CAP, DAP, and gate unit.

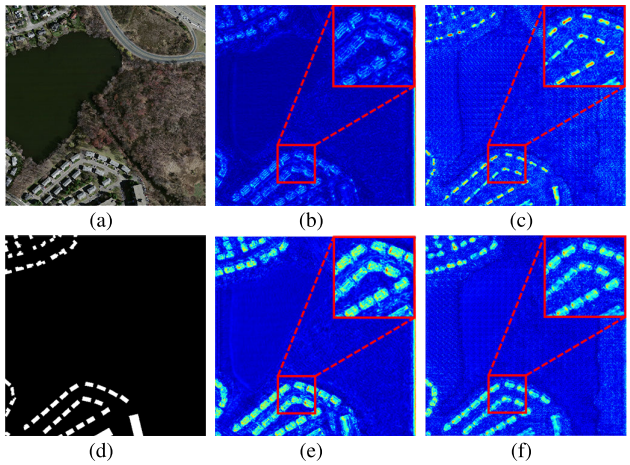


Fig. 6. Examples of DFs and SFs extracted from the CVB and TFB with the triplet loss function constraint. (a) and (d) Original image and the ground truth. (b) and (c) DFs. (e) and (f) SFs.

The former is proposed for the first time to extract complementary features from interbranch features for fusion, while the latter is used to fuse intrabranch features of different layers. As shown in Fig. 3, the CFM and MLFS are coupled with the structure that their inputs and outputs are cascaded. Moreover, the original inputs of the whole fusion scheme consist of most CNN features (i.e., $\{\mathbf{F}_i\}_{i=1,2,3,4}$) and some high-level Transformer features (i.e., $\{\mathbf{G}_j\}_{j=1,2,3,4}$). Visual comparisons of $\{\mathbf{G}_j\}_{j=1,2,\ldots,12}$ shown in Fig. 4 indicate that low-level Transformer features look bad and may not be well-learned while high-level Transformer features convey more accurate semantic and location information. This is why we choose high-level Transformer features for fusion.

1) Complementarity-Aware Fusion Module: The CFM is designed to dig out and integrate the complementary information between CNN features and Transformer features $\{\bar{\mathbf{F}}_i, \bar{\mathbf{G}}_i\}$ $(i \in \{1, 2, 3, 4\})$. Except for $\bar{\mathbf{F}}_1 = \mathbf{F}_1$ and $\bar{\mathbf{G}}_1 = \mathbf{G}_1$, other features are output from MLFS blocks. As shown in Fig. 5, the CFM introduces a CAP and two DAPs to decompose $\bar{\mathbf{F}}_i$ and $\bar{\mathbf{G}}_i$ into SFs $\bar{\mathbf{F}}_i^s$ and $\bar{\mathbf{G}}_i^s$, as well as DFs $\bar{\mathbf{F}}_i^d$ and $\bar{\mathbf{G}}_i^d$. Since complementarities are branch-specific information, DFs potentially contain complementary information. As can be seen in Fig. 5, both CAP and DAP adopt a three-convolution-layer structure as well as an additional regularization to dig out SFs and DFs. The regularization aims at pushing SFs to be similar and DFs to be dissimilar. To this end, a new triplet loss function [42] is designed for the regularization. Specifically, the triplet loss function consists of an anchor, a positive sample, and a negative sample. The optimization objective of the loss function is to minimize the distance between anchor and positive sample (i.e., make them closer) while maximizing the distance between anchor and negative sample. Here, we take $\bar{\mathbf{F}}_i^s$ as anchor, $\bar{\mathbf{G}}_i^s$ as positive sample, and $\bar{\mathbf{F}}_i^d$ as negative sample. The same operation is applied to $\bar{\mathbf{G}}_i^s$, $\bar{\mathbf{F}}_i^s$, and $\bar{\mathbf{G}}_i^d$. As a result, two triplet loss functions L_i^F and L_i^G with respect to two branches are, respectively, constructed by

$$\begin{aligned}\mathcal{L}_i^F &= \mathrm{Max}\big(0, \|\bar{\mathbf{F}}_i^s - \bar{\mathbf{G}}_i^s\|^2 - \|\bar{\mathbf{F}}_i^s - \bar{\mathbf{F}}_i^d\|^2 + m\big)\\ \mathcal{L}_i^G &= \mathrm{Max}\big(0, \|\bar{\mathbf{G}}_i^s - \bar{\mathbf{F}}_i^s\|^2 - \|\bar{\mathbf{G}}_i^s - \bar{\mathbf{G}}_i^d\|^2 + m\big)\end{aligned} \tag{6}$$

where margin value m is a hyperparameter to keep negative samples far apart. By minimize (6), DFs $\bar{\mathbf{F}}_i^d$ and $\bar{\mathbf{G}}_i^d$ are achieved and convey complementary information. The visual examples of shared and DFs demonstrated in Fig. 6 indicate that SFs are definitely similar and DFs contain complementary information (e.g., edge information and semantic information).

Next, these DFs $\bar{\mathbf{F}}_i^d$ and $\bar{\mathbf{G}}_i^d$ should be, respectively, integrated with $\bar{\mathbf{G}}_i$ and $\bar{\mathbf{F}}_i$ in a cross manner, to achieve complementary fusion. Directly stacking $\bar{\mathbf{F}}_i^d$ and $\bar{\mathbf{G}}_i$ (or $\bar{\mathbf{G}}_i^d$ and $\bar{\mathbf{F}}_i$) is rough and unsuitable since DFs not only convey useful complementary information but also contain interference information (e.g., noisy). Therefore, we design a new gate unit to filter DFs of different branches. As shown in Fig. 5, the gate unit uses a 3 × 3 convolutional layer and the sigmoid activation function to produce the two-channel building extraction maps, i.e., the building map and the nonbuilding map, under the supervision of ground truth. The building map is used as gate signals and will be multiplied with DFs. It is based on the following concept. That is, lower map values (i.e., inaccurate predictions of building) mean more interference information and lead to the suppression of interference information. As a consequence, the gate signals $\mathbf{g}_i^F$ and $\mathbf{g}_i^G$ with respect to the CVB and the TFB are, respectively, calculated by

$$\begin{aligned}\mathbf{g}_i^F &= \psi\big(\sigma\big(W_{3\times3}\big(\bar{\mathbf{F}}_i\big)\big)\big)\\ \mathbf{g}_i^G &= \psi\big(\sigma\big(W_{3\times3}\big(\bar{\mathbf{G}}_i\big)\big)\big)\end{aligned} \tag{7}$$

where $W_{3\times3}$ represents the 3 × 3 convolutional layer, σ denotes the sigmoid activation function, and ψ means the selection operation of the building map. We, respectively, multiply $\bar{\mathbf{F}}_i^d$ and $\bar{\mathbf{G}}_i^d$ by $\mathbf{g}_i^F$ and $\mathbf{g}_i^G$, to achieve two filtered DFs.

Finally, filtered DFs of current branch are used to complement the features of opposite branch and result in fused features $\mathbf{F}_i'$ and $\mathbf{G}_i'$, which is calculated by

$$\begin{aligned}\mathbf{F}_i' &= W_{3\times3}\big(\mathrm{Add}\big(\bar{\mathbf{F}}_i^s, \bar{\mathbf{F}}_i^d, \bar{\mathbf{G}}_i^d \odot \mathbf{g}_i^G\big)\big)\\ \mathbf{G}_i' &= W_{3\times3}\big(\mathrm{Add}\big(\bar{\mathbf{G}}_i^s, \bar{\mathbf{G}}_i^d, \bar{\mathbf{F}}_i^d \odot \mathbf{g}_i^F\big)\big)\end{aligned} \tag{8}$$

where $\odot$ means the Hadamard product applied to each channel of $\bar{\mathbf{G}}_i^d$ and $\bar{\mathbf{F}}_i^d$, and Add is the addition operation. The visual comparisons shown in Fig. 7 demonstrate that the proposed CFM makes fused features have more accurate building shape

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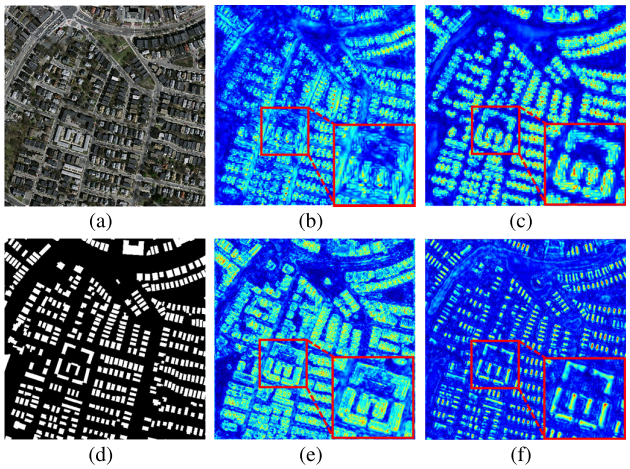


Fig. 7. Fused results of features $\mathbf{F}_4'$ and $\mathbf{G}_4'$ by CFM. (a) Original image. (b) Features $\bar{\mathbf{F}}_4$. (c) Features $\bar{\mathbf{G}}4$. (d) Ground truth. (e) Fused features $\mathbf{F}_4'$. (f) Fused features $\mathbf{G}_4'$.

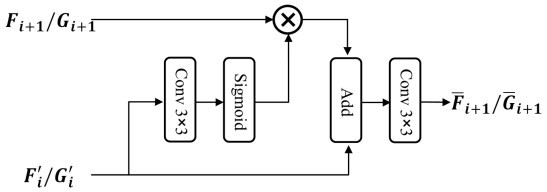


Fig. 8. Fusion scheme of different layer features in CVB and TFB. F_{i+1}/G_{i+1} is the shallow features. F_i'/G_i' is the deep features. $\bar{F}_{i+1}/\bar{G}_{i+1}$ is the fusion result.

and semantic representation. Except for $\mathbf{F}_4'$ and $\mathbf{G}_4'$ which are used for final prediction, other features $\{\mathbf{F}_i', \mathbf{G}_i'\}_{i=1,2,3}$ will be delivered to subsequent intrabranch fusion.

2) Multilayer Fusion Scheme: As known, low-level features are good at capturing high-resolution details while high-level features tend to learn semantic representation. Therefore, integrating multilayer features contributes to exploiting complementary information and improve the performance of networks. In this section, we propose an MLFS to fuse intrabranch features. Inspired by the structure of the feature pyramid network [43], we build a top-down pathway, which gradually fuse high-level and low-level features to produce highly semantic yet high-resolution features. To this end, a set of MLFS blocks are constructed to fuse $\mathbf{F}_i'$ and $\mathbf{F}_{i+1}$, as well as $\mathbf{G}_i'$ and $\mathbf{G}_{i+1}$. We introduce the gate mechanism due to the observation in Fig. 4 that shallow features contain more interference information than deep features. Therefore, during the fusion of multilayer features, shallow features should be filtered with gate signals. Here, deep features is used to guide the generation of gate signals in the MLFS blocks. The detailed structure of MLFS block is demonstrated in Fig. 8. The gate signals $\hat{\mathbf{g}}_i^F$ and $\hat{\mathbf{g}}_i^G$ of the ith MLFS block are obtained by

$$\begin{aligned}\hat{\mathbf{g}}_i^F &= \sigma\left(W_{3\times3}\left(\mathbf{F}_i'\right)\right)\\ \hat{\mathbf{g}}_i^G &= \sigma\left(W_{3\times3}\left(\mathbf{G}_i'\right)\right).\end{aligned} \tag{9}$$

As can be observed in Fig. 8, deep features and filtered shallow features are added and then fed into a 3 × 3 convolutional layer for further fusion. The output of the MLFS block is achieved by

$$\begin{aligned}\bar{\mathbf{F}}_{i+1} &= W_{3\times3}\left(\mathrm{Add}\left(\mathbf{F}_i', \mathbf{F}_{i+1}\odot\hat{\mathbf{g}}_i^F\right)\right)\\ \bar{\mathbf{G}}_{i+1} &= W_{3\times3}\left(\mathrm{Add}\left(\mathbf{G}_i', \mathbf{G}_{i+1}\odot\hat{\mathbf{g}}_i^G\right)\right)\end{aligned} \tag{10}$$

where $\mathbf{F}_i'$ and $\mathbf{G}_i'$ are the deep features from CFM, and $\mathbf{F}_{i+1}$ and $\mathbf{G}_{i+1}$ are the shallow features from two branches. Features $\bar{\mathbf{F}}_{i+1}$ and $\bar{\mathbf{G}}_{i+1}$ are then input into the next CFM.

C. Optimization Objective of CLGFF-Net

According to the above descriptions, the proposed CLGFF-Net has three types of loss functions, including the building extraction loss function $\mathcal{L}_1$, the gate unit loss function $\mathcal{L}_2$, and the triplet loss function $\mathcal{L}_3$. These loss functions are coupled to construct a joint loss function. By minimizing the joint loss function, the training of the CLGFF-Net is achieved.

To make the final building extraction results close to the ground truth, we adopt the binary cross-entropy loss to implement pixelwise binary classification ("building" and "nonbuilding"). As a result, the loss function $\mathcal{L}_1$ is described as

$$\mathcal{L}_1 = -\frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{y}_n\log\left(\hat{\mathbf{y}}_n\right)+\left(1-\mathbf{y}_n\right)\log\left(1-\hat{\mathbf{y}}_n\right)\right) \tag{11}$$

where N is the number of training samples, $\hat{\mathbf{y}}_n$ is the predicted probability value, and $\mathbf{y}_n$ is the true value. In addition, the gate units use predicted building maps as outputs, which are also supervised by the ground truth. Therefore, similar to $\mathcal{L}_1$, the gate unit loss function $\mathcal{L}_2$ is based on the binary cross-entropy function. The detailed description is given by

$$\begin{aligned}\mathcal{L}_2 &= \sum_{i=1}^{4}\left(\mathcal{L}_i^a+\mathcal{L}_i^b\right)\\ \text{s.t., } \mathcal{L}_i^a &= -\frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{y}_n\log\left(\tilde{\mathbf{y}}_n^i\right)+\left(1-\mathbf{y}_n\right)\log\left(1-\tilde{\mathbf{y}}_n^i\right)\right)\\ \mathcal{L}_i^b &= -\frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{y}_n\log\left(\check{\mathbf{y}}_n^i\right)+\left(1-\mathbf{y}_n\right)\log\left(1-\check{\mathbf{y}}_n^i\right)\right)\end{aligned} \tag{12}$$

where $\mathcal{L}_i^a$ and $\mathcal{L}_i^b$ are the gate unit loss functions with respect to the CVB and TFB, and $\tilde{\mathbf{y}}_n^i$ and $\check{\mathbf{y}}_n^i$ are the prediction results generated by the ith CFM of two different branches. After the incorporation of the triplet loss function mentioned in (8), the joint loss function can be achieved by

$$\mathcal{L} = \mathcal{L}_1+\mathcal{L}_2+\beta\mathcal{L}_3 \tag{13}$$

where $\mathcal{L}_3 = \sum_{\mathrm{i}=1}^{4}(L_i^F+L_i^G)$, and β is a tradeoff parameter between $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_3$.

IV. Experimental Results and Discussion

In this section, we first introduce the experimental setting, including the three widely-used datasets, the common evaluation metrics, and some implementation details. Next, we conduct ablation experiments to validate the effectiveness of several components in the CLGFF-Net. Finally, we compare the proposed network with both state-of-the-art semantic segmentation networks and recent building extraction methods, in terms of both the visualization and quantitative results.

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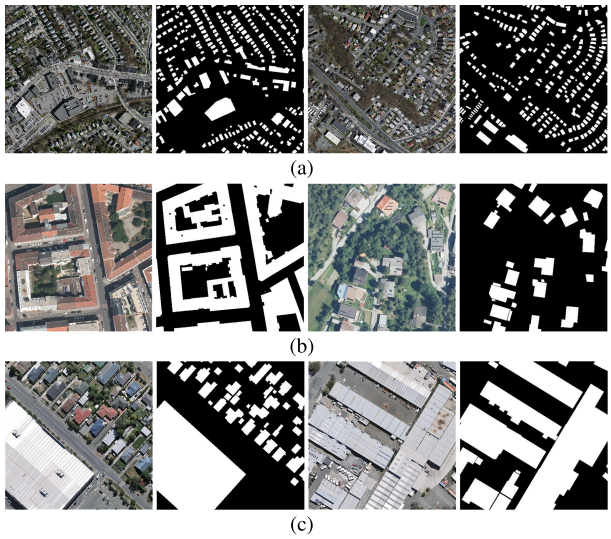


Fig. 9. Visualized examples of three well-known datasets. (a) Massachusetts Building dataset. (b) Inria Image Labeling dataset. (c) WHU Building dataset.

A. Dataset Description

In the experiment, three widely-used building extraction datasets are tested, including the Massachusetts Building dataset [44], the Inria Aerial Image Labeling dataset [45], and the WHU Building dataset [46]. Fig. 9 shows several examples of the three datasets and the related ground truth of building extraction maps. In Fig. 9, the white and black pixels represent buildings and nonbuildings, respectively.

1) Massachusetts Dataset: This dataset is a collection of 151 aerial images with 1500 × 1500 pixels, covering an area of about 340 km^2. The dataset is divided into three parts: a training set with 137 images, a test set with ten images, and a validation set with four images. In addition, the labels for the test and validation sets are manually corrected to improve the accuracy of the evaluation. In order to facilitate training, we have preprocessed each image by resizing it from an initial size of 1500 × 1500 pixels to 1536 × 1536 pixels and partitioning it into nine nonoverlapping blocks of size 512 × 512 pixels. As a result, the training set contains 1233 images, while the validation and test sets contain 36 and 90 images, respectively.

2) Inria Dataset: This dataset covers an area of 810 km^2, which consists of 360 images with a spatial resolution of 0.3 m, each sized at 5000 × 5000 pixels. The images were captured from ten different types of urban areas, including densely populated areas such as the financial district of San Francisco, CA, USA, as well as sparsely populated alpine towns like Lienz, Tyrol, Austria. Here, we partition the images in the dataset with ground truth into nonoverlapping blocks of 512 × 512 pixels and randomly divide the new dataset into new training, validation, and test sets at a ratio of 7:1:2. As a result, the newly created dataset consists of 8515 images in the training set, 1161 images in the validation set, and 2419 images in the test set.

3) WHU Dataset: This dataset includes 187 000 buildings with 0.3-m ground resolution, which covers the 450-km^2 area in Christchurch, New Zealand. The dataset is composed of 8189 images sized 512 × 512. In detail, the training set of the WHU Building dataset contains 4736 images with a total of 130 500 buildings, the validation set contains 1036 images with a total of 14 500 buildings, and the test set contains 2416 images with a total of 42 000 buildings.

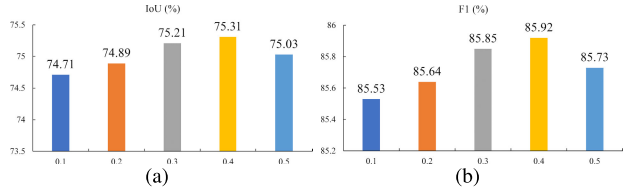


Fig. 10. Influences of the parameter β on the building extraction performance of the proposed method in terms of IoU and F1. (a) IoU. (b) F1.

B. Experimental Setting

Following the work in [47], our method adopts some common data augmentation operations. These operations include random resize, random crop, random rotation, and random horizontal flip, where the ratio of random resize is between 0.5 and 2.0, and the size of random crop is 512 × 512. In addition, we adopt a cyclic learning rate decay schedule [48] that makes the learning rate increase first and then decrease. The Adam [49] optimizer with a base learning rate $\gamma = 10^{-3}$ is applied here. For the three datasets, the entire training process includes 300 epochs, and the batch size is set to 4. In the experiments, one hyperparameter m in the triplet loss function is set as the default value following the suggested configuration in [42]. The proposed network in this article is implemented with PyTorch. All the experiments are performed on a server equipped with an Intel Xeon[1] Gold 6148 CPU and an Nvidia GeForce GTX 3090 GPU.

In addition, the effect of hyperparameter β in (13) will be discussed here, which is used to balance the triplet loss. In order to find which set of hyperparameters provides the best performance, we conduct a series of experiments. In these experiments, we keep other parameters and settings constant and set the β to different values such as 0.1–0.5. The quantitative results for different β are recorded in Fig. 10. The proposed method achieves optimal performance when β is set to 0.4. Therefore, the hyperparameter β is fixed in all experiments for the training of CLGFF-Net.

In order to evaluate the performance of our proposed network and other state-of-the-art methods, we adopt four widely-used evaluation metrics, i.e., recall (R), precision (P), intersection over union (IoU), and F1 score (F1). In general, the higher the objective metrics are, the better the performance of the learning model. Among these metrics, R is the percentage of building areas extracted by the model compared to the total area of existing buildings. The higher the R is, the better the model can capture the actual buildings present, and thus, the less likely it would miss building areas. P is the percentage of correctly predicted buildings. A higher P implies that more building pixels are correctly predicted as building areas. IoU measures the degree of overlap between the

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TABLE I
EFFECTS OF DIFFERENT COMPONENTS, INCLUDING THE CVB, TFB, CFM, AND MLFS, ON THE CLGFF-NET OVER THE MASSACHUSETTS BUILDING DATASET (%)

Methods	CVB	TFB	CFM	MLFS	R	P	IoU	F1
CVB	√				84.93	82.06	71.63	83.47
TFB		√			85.87	83.91	73.73	84.88
CVB + TFB	√	√			85.47	84.66	74.01	85.06
CVB + TFB + CFM	√	√	√		86.38	84.74	74.76	85.55
CVB + TFB + MLFS	√	√		√	**86.92**	84.42	74.90	85.65
CLGFF-Net(ours)	√	√	√	√	86.80	**85.05**	**75.31**	**85.92**

predicted building areas and the actual building areas. Higher IoU represents that the predicted building area better matches the actual building area. F1 combines the tradeoffs between R and P, providing the harmonic average of these two metrics. A higher F1 indicates that the model can correctly classify buildings while extracting as much of the actual building area as possible, striking a good balance between R and P.

C. Ablation Studies

In order to analyze the contribution of components in the proposed CLGFF-Net, we conduct a series of ablation experiments here. First, only the CVB or TFB is used in the feature extraction process, and then, the feature maps of the last four stages are fused in a feature pyramid manner [43] to generate the building extraction results. Second, only the CVB and TFB are utilized (denoted as CVB + TFB) and the feature maps obtained from the CVB and TFB are concatenated and further learned by convolution before mapping to building masks. Third, the CFM is incorporated into CVB and TFB (denoted as CVB + TFB + CFM). That is, the feature maps from different layers of two branches are integrated in the CFM. In addition, the CFM is removed and the MLFS is introduced for the CVB and TFB features (denoted as CVB + TFB + MLFS). Finally, all the CVB, TFB, CFM, and MLFS are incorporated to constitute our CLGFF-Net.

The numerical results of ablation studies are shown in Table I. It can be observed that directly combining the CVB with TFB can enhance the building extraction performance when compared to using the CVB or TFB alone. According to Table I, the critical IoU and F1 of CVB + TFB + CFM are both higher than those of CVB + TFB. The increases of IoU and F1 are, respectively, reached to 0.75% and 0.49%. In addition, the CVB + TFB + MLFS leads to an increase in IoU by 0.89% and F1 by 0.59%, as compared with CVB + TFB. What is more important, the IoU and F1 can be further improved by simultaneously using the CVB, TFB, CFM, and MLFS (CLGFF-Net).

The visualization results by the inclusion and exclusion of CFM or MLFS are shown in Figs. 11 and 12, respectively. In the red boxes of the first and third rows in Fig. 11, the w/o CFM approach fails to identify the buildings and incorrectly classifies gaps between buildings as buildings. By contrast, the CLGFF-Net with CFM demonstrates good performance in extracting buildings with complicated contours or occluded by trees and shadows. It certainly proves that the CFM strengthens the anti-interference ability of the proposed network. In addition, the visualization results in Fig. 12 exhibit

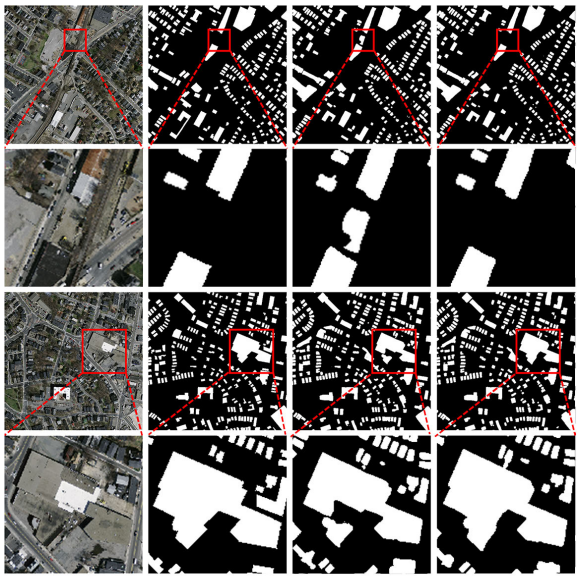


Fig. 11. Examples of ablation experiments to test the effect of CFM on the Massachusetts dataset. (a) Original image. (b) Ground truth. (c) Building extraction map generated by w/o CFM (i.e., the CLGFF-Net without the CFM). (d) Building extraction map generated by the CLGFF-Net. The even rows are the enlargements of the areas in the red boxes of the odd rows.

the performance comparison when adding and removing the MLFS. Obviously, the lack of MLFS leads to incomplete and missing building extraction, as well as significant differences in shape between predicted and actual buildings in RSIs.

D. Comparison With Other Methods

To evaluate the performance of the proposed network, we compare it with both classical and state-of-the-art methods on the three publicly available datasets. The selected methods maintain PSPNet [11], DeepLabv3+ [12], HRNet [13], ConvNeXt [14], Swin-Transformer [26], Next-ViT [27], and UniFormer [28]. Among them, the PSPNet, DeepLabv3+, HRNet, and ConvNeXt are convolutional networks, while Swin-Transformer relies on Transformer structure. Considering this work is the fusion of both convolution and Transformer, and fusion networks (i.e., Next-ViT and UniFormer) are also compared. Brief descriptions of comparison results are given in the following.

1) PSPNet: This method proposes a pyramid pooling module, which achieves promising performance in tasks such as scene parsing by employing subregions of multiple sizes to capture scene features from different levels.

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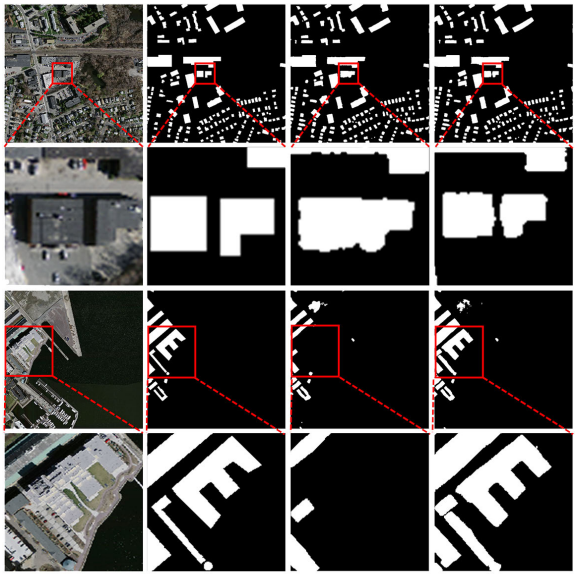


Fig. 12. Examples of ablation experiments to test the effect of MLFS on the Massachusetts dataset. (a) Original image. (b) Ground truth. (c) Building extraction map generated by w/o MLFS (i.e., the CLGFF-Net without the MLFS). (d) Building extraction map generated by the CLGFF-Net. The even rows are the enlargements of the areas in the red boxes of the odd rows.

2) DeepLabv3+: This method employs atrous convolution to expand the receptive field, incorporating a larger range of context information. Additionally, it proposes an atrous spatial pyramid pooling module that captures multiscale context information without increasing the number of parameters.

3) HRNet: This method achieves the preservation of high-resolution representation by connecting high-resolution to low-resolution feature maps in parallel and repeating the multiresolution information exchange. This design enables HRNet to take full advantage of multiscale features.

4) ConvNeXt: This method is a powerful CNN with a structure similar to ResNet [50]. By borrowing the design principles of ViT, ConvNeXt exhibits performance comparable to or even superior to Transformer in many cases.

5) Swin-Transformer: This method introduces a hierarchical Transformer structure that employs the mechanism of shifted windows to restrict self-attention computation to nonoverlapping local windows, thus improving the locality of the Transformer.

6) Next-ViT: This method is an efficient CNN-Transformer hybrid network architecture that combines the CNN and Transformer in a serial way to integrate the local context and global long-range dependencies.

7) UniFormer: This method combines the advantages of convolution and self-attention in the Transformer architecture to significantly improve the efficiency and performance of image recognition and segmentation.

First, the quantitative analysis results on three datasets are analyzed here. Table II presents the objective metrics tested on the small dataset, i.e., Massachusetts Building dataset. Among all the compared methods, PSPNet obtains the most unsatisfied results, with the R of 78.07%, P of 84.41%, IoU of 68.23%, and F1 of 81.12%. Due to the ability of Transformer in exploiting attention mechanism and capturing global long-range dependencies, the Transformer-based approach (Swin-Transformer) outperforms the CNN-based approaches (PSPNet, DeepLabv3+, HRNet, and ConvNeXt). Notably, by borrowing the design principles of ViT, ConvNeXt also achieves competitive results with Swin-Transformer. By combining convolution and Transformer in one fusion method, the advantages of both networks/structures can be utilized for better prediction results. Though the UniFormer and NextViT fuse convolution and Transformer in one framework, they do not fully explore and fuse the complementary information in interbranch or intrabranch to guide more effective fusion scheme. As a result, limited performance improvement will be obtained by these two fusion networks. In contrast, our approach proposed CFM to effectively explore and fuse complementary information between CVB and TFB features, resulting in superior performance. In general, though our method does not achieve the highest score on P, it outperforms the compared methods in the other three metrics. One main reason is that our approach can extract buildings more completely and effectively reduce problems such as building extraction omission. However, this advantage also results in more nonbuilding pixels being incorrectly predicted as building pixels, thereby leading to a decrease of P. Moreover, the metrics of R and P usually present a trend of "when one is rising, the other is falling." In order to evaluate the overall performance of these methods, the IoU and F1 can measure the method performance from different aspects. In consistent with the results in Table II, the results of the other two datasets (the Inria Aerial Image Labeling dataset and the WHU Building dataset) also demonstrate the effectiveness of the proposed method in performance improvements, as shown in Tables III and IV. Compared to the dataset of Massachusetts, these two datasets have larger data volumes and more clear building labels with well-defined boundaries. It can be seen from Table III that both the proposed method and the compared state-of-the-art methods achieve competitive results. It still can be noticed that the proposed approach can obtain varying degrees of performance improvement in terms of R, IoU, and F1 for these two datasets.

Then, several examples of visual comparison results are shown here. In Fig. 13, there are two main common characters for the images in these examples of Massachusetts Building test dataset. That is, the buildings are densely located and they have a very similar color/texture to the background.

TABLE II
COMPARISON OF QUANTITATIVE METRICS FOR THE MASSACHUSETTS BUILDING DATASET (%)

Methods	R	P	IoU	F1
PSPNet [11]	78.07	84.41	68.23	81.12
DeepLabv3+ [12]	80.65	87.53	72.34	83.95
HRNet [13]	80.14	85.24	70.37	82.61
ConvNeXt [14]	81.72	87.33	73.06	84.43
Swin-Transformer [26]	82.24	**87.64**	73.69	84.85
UniFormer [28]	82.41	87.60	73.80	84.92
Next-ViT [27]	81.24	86.57	72.15	83.82
CLGFF-Net(ours)	**86.85**	85.03	**75.33**	**85.93**

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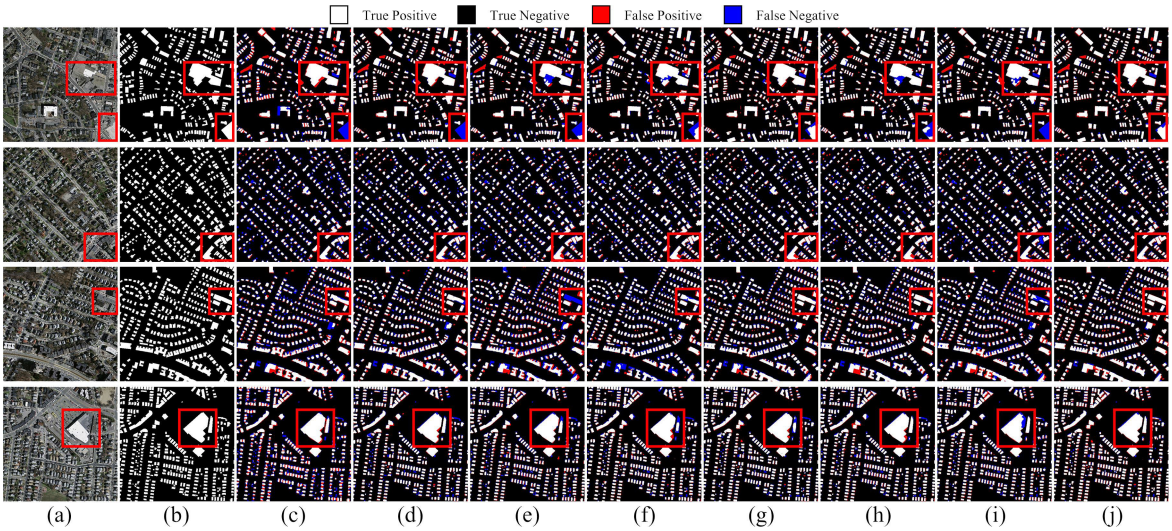


Fig. 13. Building extraction examples of different methods on the Massachusetts Building test dataset. (a) Original image. (b) Ground truth. (c) PSPNet. (d) DeepLabv3+. (e) HRNet. (f) ConvNeXt. (g) Swin-Transformer. (h) UniFormer. (i) Next-ViT. (j) CLGFF-Net.

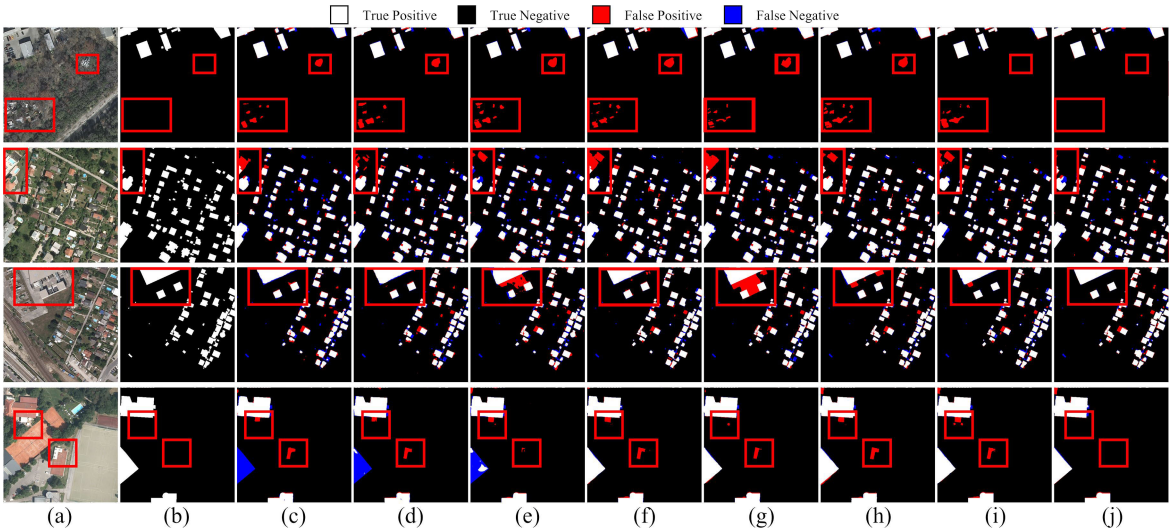


Fig. 14. Building extraction examples of different methods on the Inria Aerial Image Labeling test dataset. (a) Original image. (b) Ground truth. (c) PSPNet. (d) DeepLabv3+. (e) HRNet. (f) ConvNeXt. (g) Swin-Transformer. (h) UniFormer. (i) Next-ViT. (j) CLGFF-Net.

Thus, it is difficult to extract the buildings in these examples. In contrast, the proposed CLGFF-Net performs better than the compared methods in these examples, which can generate a better result with more buildings correctly extracted while obtaining less wrong predictions. This is mainly due to the following twofold aspects. On the one hand, we introduced an effective fusion design to jointly exploit the advantages of convolution and Transformer, which can push the network to learn more discriminative features. That is, the difference between represented features of buildings and nonbuildings is bigger, leading to higher discrimination between them even they look similar in original image space. On the other hand, the proposed fusion strategy across multiple layers will contribute to better characterization of buildings of different scales. In this manner, the densely located buildings with various shapes and sizes can be more accurately extracted by the proposed method. In addition, more visual results of Inria dataset are also shown in Fig. 14. For example, in the first two rows, the surrounding background of the building is very complex and various degrees of errors occur in the compared methods. In contrast, the visualization results indicate that the proposed method can better discriminate the buildings and surrounding background.

TABLE III
COMPARISON OF QUANTITATIVE METRICS FOR THE INRIA AERIAL IMAGE LABELING DATASET (%)

Methods	R	P	IoU	F1
PSPNet [11]	88.9	90.27	81.12	89.58
DeepLabv3+ [12]	88.46	90.78	81.17	89.61
HRNet [13]	89.24	90.20	81.35	89.72
ConvNeXt [14]	90.99	91.80	84.16	91.40
Swin-Transformer [26]	90.75	91.75	83.90	91.25
UniFormer [28]	91.21	**91.84**	84.37	91.52
Next-ViT [27]	91.21	91.76	84.31	91.49
CLGFF-Net(ours)	**92.12**	91.59	**84.94**	**91.85**

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TABLE IV
COMPARISON OF QUANTITATIVE METRICS FOR THE WHU BUILDING DATASET (%)

Methods	R	P	IoU	F1
PSPNet [11]	93.58	94.43	88.68	94.00
DeepLabv3+ [12]	93.98	95.24	89.76	94.61
HRNet [13]	94.28	95.01	89.83	94.64
ConvNeXt [14]	94.83	**95.64**	90.90	95.23
Swin-Transformer [26]	95.09	94.85	90.42	94.97
UniFormer [28]	95.07	95.01	90.55	95.04
Next-ViT [27]	95.06	95.30	90.81	95.18
CLGFF-Net(ours)	**95.89**	95.01	**91.30**	**95.45**

TABLE V
COMPARISON WITH THE MOST RECENT BUILDING EXTRACTION METHODS ON THE WHU BUILDING DATASET (%)

Methods	R	P	IoU	F1
CFENet [51]	91.56	93.70	87.22	92.62
MultiBuildNet [52]	94.06	93.50	88.29	93.78
DR-Net [53]	93.30	94.30	88.30	93.80
SiU-Net [46]	93.90	93.80	88.40	93.85
B-FGC-Net [54]	94.49	95.03	90.04	94.76
BOMSC-Net [55]	94.50	**95.14**	90.15	94.80
LCS [56]	93.90	95.38	90.71	95.12
MAP-Net [17]	94.81	95.62	90.86	95.21
CLGFF-Net(ours)	**95.89**	95.01	**91.30**	**95.45**

Finally, to further evaluate the performance of CLGFF-Net, we also compare it with some recent building extraction methods based on WHU dataset. The compared methods include CFE-Net [51], MultiBuildNet [52], DR-Net [53], SiU-Net [46], B-FGC-Net [54], BOMSC-Net [55], LCS [56], and MAP-Net [17]. Here, CFE-Net utilizes low-level features to capture building outline information and fuses high-level semantic features to extract building category information. MultiBuildNet builds a multitask learning framework within which building detection, footprint segmentation, and edge extraction tasks are achieved simultaneously. DR-Net adopts the encoder–decoder backbone of DeepLabv3+ [12] and proposes a novel architecture that combines densely connected CNNs with residual network structures. SiU-Net uses multiscale inputs with two weight-shared UNets to optimize the segmentation results when dealing with the task of building extraction. B-FGC-Net proposes a global feature information awareness module to effectively capture multiscale context information and introduces a cross-level feature recalibration module to fuse information between low- and high-level features. In BOMSC-Net, a pooling operation and a graph reasoning module are used for context inference, while directional information is introduced to guide the fine segmentation of building boundaries. The LCS proposed a line segment collaborate segmentation framework, leveraging the complementary strengths of semantic segmentation and vector extraction. This strategy effectively addresses irregular boundary challenges in architectural extraction, where semantic segmentation guides vector extraction location, and vector extraction provides precise positioning for semantic segmentation. The MAP-Net adopted a multiple parallel path learning strategy, where high-level semantic features with fixed resolution is gradually generated in each path. It can obtain spatial localization-preserved multiscale features, making the network accurately extracting tiny buildings. This WHU dataset has a fixed image size and standard experimental setting (e.g., training set, validation set, and test set). Following the works in MAP-Net [17] and BCTNet [29], we compare the results in related works, instead of reproducing the experiments. The quantitative analysis results for the WHU Building dataset are given in Table V, where the highest scores for each metric are marked in bold. Among all the compared methods, CFE-Net has the lowest P (93.7%), R (91.56%), IoU (87.22%), and F1 (92.62%). By optimizing the network architectures with the multitask learning framework, residual structures, and Siamese networks, the MultiBuildNet, DR-Net, and SiU-Net achieve better performance than CFE-Net, leading to a 2.5%, 1.74%, and 2.34% improvement in R, 1.07%, 1.08%, and 1.18% in IoU, and 1.16%, 1.18%, and 1.23% in F1, respectively. In addition, two very new methods, i.e., the LCS and MAP-Net, show better results than the above methods. Thanks to the effective fusion of local and global features, the proposed CLGFF-Net method is superior to the LCS and MAP-Net, and achieve the highest values for the most of metrics, including the R, IoU and F1.

TABLE VI
COMPARISON OF COMPLEXITY WITH THE STATE-OF-THE-ART METHODS ON THE MASSACHUSETTS BUILDING DATASET

Methods	Params(M)	FLOPs(G)	IoU(%)	F1(%)
PSPNet [11]	67.95	256.62	68.23	81.12
DeepLabv3+ [12]	62.57	254.53	72.34	83.95
HRNet [13]	65.85	93.68	70.37	82.61
ConvNeXt [14]	120.81	291.24	73.06	84.43
Swin-Transformer [26]	119.99	296.04	73.69	84.85
Uniformer [28]	79.64	254.62	73.80	84.92
Next-ViT [27]	78.2	253.84	72.15	83.82
CLGFF-Net(ours)	**136.07**	**362.23**	**75.31**	**85.92**

E. Complexity Analysis

In this section, we compare the trainable parameter (Params), floating-point operations (FLOPs), IoU, and F1 with other state-of-the-art methods on the Massachusetts Building dataset. The quantitative results are reported in Table VI. According to the results, the proposed CLGFF-Net has the largest values of Params and FLOPs, which indicates the higher complexity of our method. It is because our method adopts two branches instead of one branch which is widely-used in those comparison methods (i.e., the PSPNet, DeepLabv3+, HRNet, ConvNeXt, Swin-Transformer, UniFormer, and Next-ViT). Obviously, two branches contain much more parameters than one branch. Although our method is more complex than other methods, the gap of Params and FLOPs is only (15.26M and 70.99G, and 16.08M and 66.19G) as compared with the well-known ConvNeXt and Swin-Transformer. Furthermore, the IoU and F1 of the CLGFF-Net is larger than other methods (more than 1.51% for IoU, and 1.0% for F1), which means that added parameters contribute to the performance improvement.

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V. Conclusion

In this article, we propose a novel network architecture called the CLGFF-Net. In CLGFF-Net, the extraction of local patterns and global long-range dependencies is achieved by designing parallel CVB and TFB. In order to effectively fuse the feature maps of these two branches, the CFM is proposed to achieve a complementary fusion of these two types of features. Meanwhile, MLFS is introduced to establish cross-layer connections in multilayer feature maps. It enables fine extraction of buildings by integrating the semantic information of deep features and the detailed information of shallow features. Comprehensive experiments are conducted on three widely-used benchmark datasets: the Massachusetts Building dataset, the Inria Aerial Image Labeling dataset, and the WHU Building dataset. The quantitative and qualitative comparison results show that our CLGFF-Net achieves superior performance on three datasets. It is confirmed that our method has great potential for building extraction.

In this work, the proposed method is only performed in the building extraction task. In future work, we will extend the fusion modules to the general semantic segmentation tasks.

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